

AKHILESH BHASKAR KOTEGAR

Software Engineer | AI & Full Stack Developer



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Profile

Software Engineer and master's student at TH Deggendorf with 1.5+ years of industry experience in full-stack development and cloud infrastructure. Hands-on experience in building AI-Integrated applications using LLMs and deep learning. National hackathon winner with a passion for combining modern web technologies with AI to solve real-world problems. Actively seeking working student or internship opportunities in software engineering and AI.

Professional Experience

11/2022 – 07/2025

Bengaluru, India

Celebto Technologies Pvt Ltd, *Software Engineer Intern*

- Implemented customised Google OAuth authentication via Firebase & Kotlin, extending the default auth flow to capture specific user data for business requirements.
- Delivered pixel-perfect Next.js components from Figma designs, shipped directly to production within sprint cycles.
- Architected GCP cloud automation with RBAC security, supporting infrastructure for 1.5+ years of product scaling.
- Reduced deployment errors significantly by building Docker-based CI/CD pipelines in an Agile Scrum environment.

Education

10/2025 – Present

Deggendorf, Germany

M.Sc. Applied Computer Science, *Technische Hochschule Deggendorf* 

Focus: Software Architecture, System Design, Advanced Programming, AI & Machine Learning.

12/2021 – 07/2025

Bengaluru, India

Bachelor of Engineering (B.E.) in Information Science and Engineering,

Ramaiah Institute of Technology

- GPA: 8.05 / 10
- Thesis: Anomaly Detection in Space Logs using Self-Supervised Learning (DBLCL).

Projects

04/2026 – 05/2026

Synapse-Query , *Production-grade RAG System for Technical Document Analysis.*

- **Architected a Hybrid Retrieval Pipeline:** Combined dense vector search (ChromaDB) with BM25 via **Reciprocal Rank Fusion (RRF)** and Cross-Encoder re-ranking for high-precision document retrieval.
- **Optimized for Production Efficiency:** Deployed models using **ONNX Runtime** to eliminate PyTorch dependencies, significantly reducing memory overhead and inference latency.
- **Engineered Robust Data Ingestion:** Built an automated **OCR fallback system** (Tesseract) and citation-resolution logic to ensure high-fidelity extraction from complex technical PDFs.

02/2026 – 04/2026	PITWALL - F1 Immersive Dashboard  <i>Real-time F1 telemetry dashboard with 3D car viewer and race replay</i> • Developed full-stack F1 telemetry dashboard streaming real-time race replay for 20 drivers via Server-Sent Events at 10 fps with live speed, DRS, gear, and tyre data. • Built Python data pipeline using FastF1, NumPy, and Pandas to process raw 20 Hz telemetry; implemented in-memory cache to eliminate reprocessing of multi-hundred MB race sessions. • Rendered interactive 3D F1 car in-browser using React Three Fibre with team-colour-matched dynamic lighting across dark and light modes. • Deployed frontend on Vercel and Dockerized FastAPI backend on Railway; supports full race, qualifying, and sprint results across all F1 seasons from 1950 to 2026.
12/2025 – 01/2026	Allergy Detector  <i>Next.js 16, TypeScript, Prisma, PostgreSQL, Google Gemini API</i> • Engineered a 3-Tier AI Analysis Pipeline: Built a fail-safe engine using Gemini, Scikit-learn, and rule-based logic to ensure continuous operation and graceful degradation. • Eliminated AI Hallucinations: Integrated Google Search-grounded LLMs with the Open Food Facts API to verify safe product alternatives against real-world data. • Implemented Real-Time Online Learning: Utilized Scikit-learn’s partial_fit to enable instant model updates from user feedback, bypassing the need for full retraining.

Awards	
20/12/2023	1st Prize – HackZion National Hackathon,, <i>AMC Engineering College, Bangalore</i> • Secured 1st Place (National Level) for building "HealthGuard, An AI-powered nutrition platform using Large Language Models (LLMs) for natural language analysis.
11/11/2023	3rd Prize — Illusive Alphabits, <i>Ramaiah Institute of Technology</i> • Placed 3rd in a competitive Linux & debugging challenge involving terminal command decoding, error fixing and hidden code discovery across multiple levels

Skills	
Web Development & Frameworks Next.js 16, React.js, TypeScript, Node.js, Express, Tailwind CSS, Shadcn/UI, HTML5/CSS3	Backend, Database & Cloud PostgreSQL, MongoDB, Prisma ORM, AWS (Lambda, API Gateway), Vercel, Docker, REST APIs.
AI & Data Science Python, TensorFlow, Keras, Google Gemini API, Anthropic Claude API, LLMs, Pandas, NumPy.	Mobile & Other Languages Java, Kotlin, Flutter
Tools & DevOps Git, GitHub, CI/CD Pipelines, Postman, Figma.	Languages English (C1 — Professional), German (A2 — Learning), Kannada (Native)